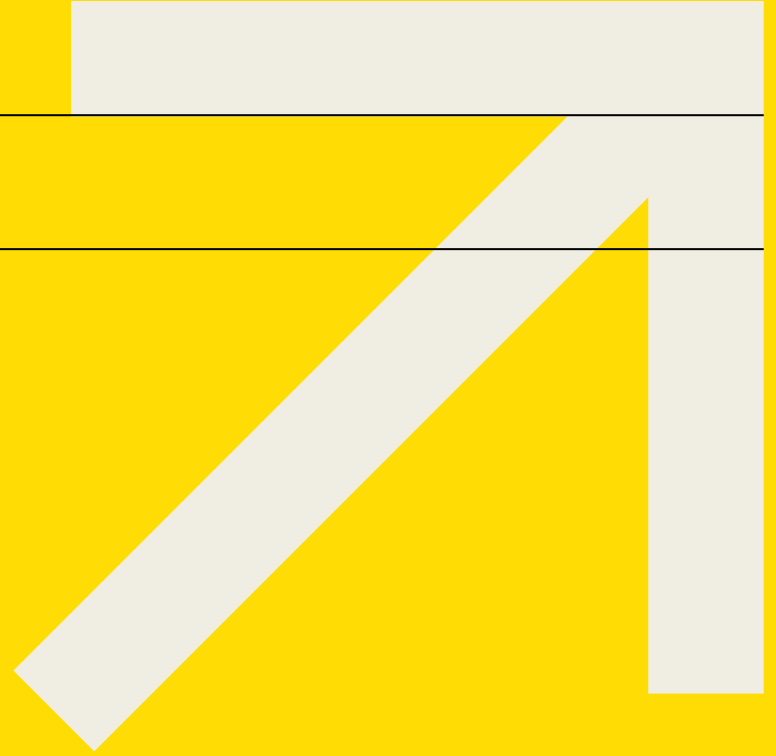


Week 3

Progress



BIG ISSUE



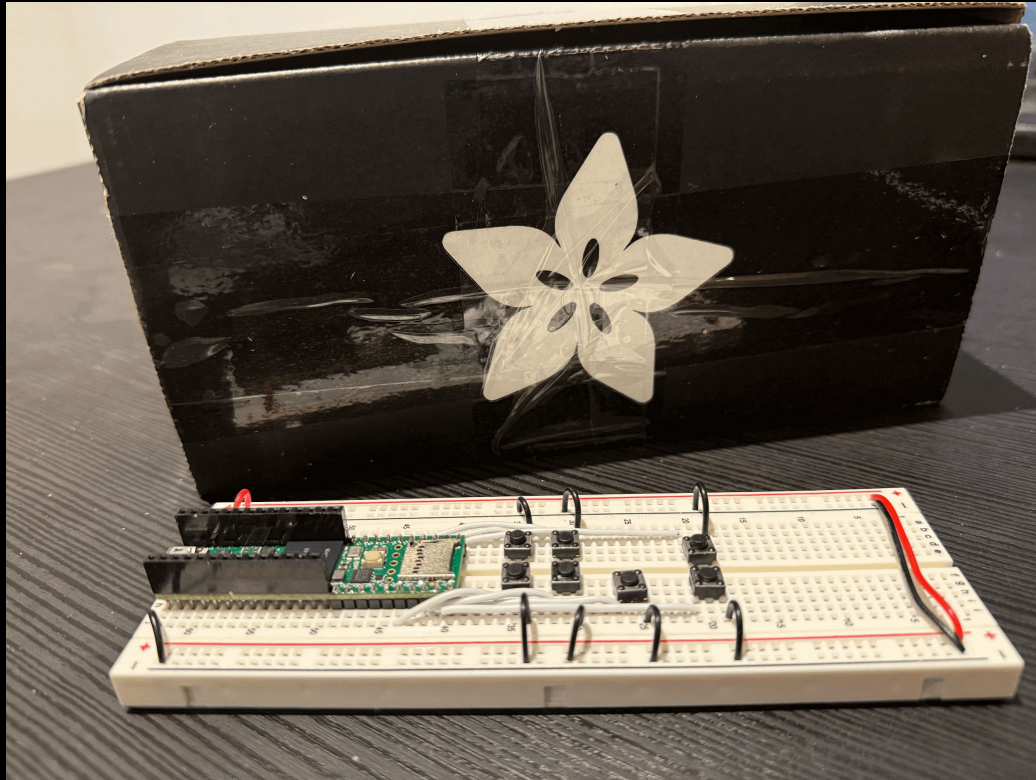
BIG ISSUE

RaspberryPi

BIG ISSUE



Our Good Friend The Teensy



Created Best Named C++ Code Doc Of All Time(Pointer I Hardly Know Her)

PointerIHardlyKnowHerino

```
1 int buttons[7] = { 28, 30, 31, 32, 33, 35, 37 };
2 bool buttonState[7] = { HIGH, HIGH, HIGH, HIGH, HIGH, HIGH, HIGH };
3 bool lastButtonState[7] = { HIGH, HIGH, HIGH, HIGH, HIGH, HIGH, HIGH };
4
5 bool wav = false;
6 bool env = false;
7
8 /*
9 Pin 28 is Snap 0 Pin 30 is Reset 1
10 Pin 32 is Snpl 3 Pin 31 is DownInc 2
11 Pin 35 is Wav 5 Pin 37 is Env 6
12 Pin 33 is UpInc 4
13 */
14
15 void setup() {
16   Serial.begin(9600);
17   for (int i = 0; i < 7; i++) {
18     pinMode(buttons[i], INPUT_PULLUP);
19   }
20 }
21
22 void loop() {
23   setWav();
24   setEnv();
25   checkButtons();
26   if(buttonState[0] == LOW && lastButtonState[0] == HIGH) {
27     snap();
28   }
29   if(buttonState[1] == LOW && lastButtonState[1] == HIGH) {
30     clear();
31   }
32   if(buttonState[3] == LOW && lastButtonState[3] == HIGH) {
33     Keyboard.press('G');//snp
34     delay(100);
35     Keyboard.release('G');
36   }
37   if(buttonState[2] == LOW && lastButtonState[2] == HIGH) {
38     downInc();
39   }
40   if(buttonState[4] == LOW && lastButtonState[4] == HIGH) {
41     upInc();
42   }
43 }
44
45 void checkButtons() {
46   for (int i = 0; i < 7; i++) {
47     lastButtonState[i] = buttonState[i];
48     buttonState[i] = digitalRead(buttons[i]);
49   }
50 }
51
52 void snap(){
53   if (wav == true) {
54     Keyboard.press('e');//wavsnap
55     delay(100);
56     wav = false;
57     Keyboard.release('e');
58   } else if (env == true) {
59     Keyboard.press('f');//envsnap
60     delay(100);
61     env = false;
62     Keyboard.release('f');
63   }
64 }
65
66
```

```
67 void clear(){
68   if (wav == true) {
69     Keyboard.press('m');//reset wav
70     delay(100);
71     wav = false;
72     Keyboard.release('m');
73   } else if (env == true) {
74     Keyboard.press('a');//reset env
75     delay(100);
76     env = false;
77     Keyboard.release('a');
78   }
79 }
80
81 void upInc(){
82   if (wav == true) {
83     Keyboard.press('S');//inc wav
84     delay(100);
85     wav = false;
86     Keyboard.release('S');
87   } else if (env == true) {
88     Keyboard.press('t');//inc env
89     delay(100);
90     env = false;
91     Keyboard.release('t');
92   }
93 }
94
95 void downInc(){
96   if (wav == true) {
97     Keyboard.press('b');//dec wav
98     delay(100);
99     wav = false;
100     Keyboard.release('b');
101   } else if (env == true) {
102     Keyboard.press('t');//dec env
103     delay(100);
104     env = false;
105     Keyboard.release('t');
106   }
107 }
108
109 void setWav() {
110   if (buttonState[5] == LOW && lastButtonState[5] == HIGH) {
111     wav = true;
112     env = false;
113     Serial.print("wav");
114   }
115 }
116
117 void setEnv() {
118   if (buttonState[6] == LOW && lastButtonState[6] == HIGH) {
119     env = true;
120     wav = false;
121     Serial.print("env");
122   }
123 }
124
125
```

Created Test Mode and ASCII Based Device Mode

```
void keyPressed() { //test true is for keyboard control test false is obscure ascii to talk to teensy
  if ((test == true && key == 'w') || (test == false && key == '€')) {
    wavSnap();
    println("wavSnap");
  } else if ((test == true && key == 'e') || (test == false && key == 'f')) {
    envSnap();
    println("envSnap");
  } else if ((test == true && key == '[') || (test == false && key == '†')) {
    decEnvMult();
    println("decEnv");
  } else if ((test == true && key == ']') || (test == false && key == '‡')) {
    incEnvMult();
    println("incEnv");
  } else if ((test == true && key == '-') || (test == false && key == '‰')) {
    decWavMult();
    println("decWav");
  } else if ((test == true && key == '=') || (test == false && key == 'Š')) {
    incWavMult();
    println("incWav");
  } else if ((test == true && key == 's') || (test == false && key == '€')) {
    smpl();
    println("smpl");
  } else if ((test == true && key == 'r') || (test == false && key == '™')) {
    resetWav();
    println("resetWav");
  } else if ((test == true && key == 't') || (test == false && key == '¤')) {
    resetEnv();
    println("resetEnv");
  } else if (key == (char)27) { //Escape key always exits program
    clearCache();
    println("Program Exit");
    exit();
  }
}
```

Reimplemented Reset Code

```
void resetWav() {
    wavMult=1;
    clearWav = true;
    filereset.save("wav_image.jpg");
}

void resetEnv() {
    envMult =1;
    clearEnv = true;
    filereset.save("env_image.jpg");
}

void smpl() {
    float[] wav;
    if (clearWav) {
        wav = resetSin;
    } else {
        wav = processWavImage();
    }
    AudioSample smpl = createSample(wav);
    playWav(smpl);
    if (clearEnv) {
        playDemoEnv(smpl);
    } else {
        playEnv(smpl);
    }
}
```

```
PImage filereset = createImage(camwidth, camheight, RGB);

float[] genSinWav() { //create sin wav arr
    float[] sinewave = new float[640];

    for (int i = 0; i < 640; i++) {
        sinewave[i] = sin(TWO_PI*i/640);
    }
    return sinewave;
}

float[] genBasicEnv() { //create reset env
    float attackTime = 0.1;
    float sustainTime = 0.5;
    float releaseTime = 0.4;
    return new float[]{attackTime, sustainTime, releaseTime};
}

void clearCache() {
    resetWav();
    resetEnv();
    println("Image Cache Cleared");
}
```

Lots And Lots Of UI(Mapping Code)

```
float volume = 0.2;
float mode = 0.0; //0.0 is Wav Mode 1.0 is Env Mode 2.0 is IMG mode

float[] mappedFlArr(float[] flArr) { //Map wav values to fit the ui screen
    float[] temp = new float[flArr.length];
    for (int i = 0; i < flArr.length; i++) {
        temp[i] = map(flArr[i], -1, 1, 40, 320);
    }
    return temp;
}

float[] mapEnv(float[] env) { //Map envelope values to fit the ui screen
    float totalTime = env[0]+env[1]+env[2];
    float attackTime = map(env[0], 0, totalTime, 0, 640);
    float sustainTime = map(env[0], 0, totalTime, 0, 640);
    float releaseTime = map(env[0], 0, totalTime, 0, 640);
    return new float[]{attackTime, sustainTime, releaseTime};
}
```

Lots And Lots Of UI("Display" Code)

```
void drawScreen() { //draw blank screen
  fill(255);
  stroke(0);
  strokeWeight(2);
  rect(40, 20, 640, 320);
}

void drawScreenBox() { //draw screen outline
  stroke(0);
  strokeWeight(2);
  noFill();
  rect(40, 20, 640, 320);
}

void drawWav(float[] wavArr) { //takes wav arr draws it
  stroke(225, 0, 0);
  strokeWeight(2);
  float[] wav = mappedFlArr(wavArr);
  for (int i = 0; i < wav.length-1; i++) {
    line(i+40, wav[i], i+41, wav[i+1]);
  }
}

void drawEnv(float[] envArr) { //takes env arr draws it
  stroke(225, 0, 0);
  strokeWeight(2);
  float[] env = mapEnv(envArr);
  line(40, 340, 40+env[0], 200);
  line(40+env[0], 200, 40+env[0]+env[1], 200);
  line(40+env[0]+env[1], 200, 680, 340);
}
```

Lots And Lots Of UI(Pre Graphics Design)



Lots And Lots Of UI(Illustrator Designs)

Smpl A Sketch

Mode

Vol.

Stylized Text



Icons/Buttons

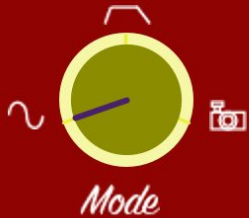
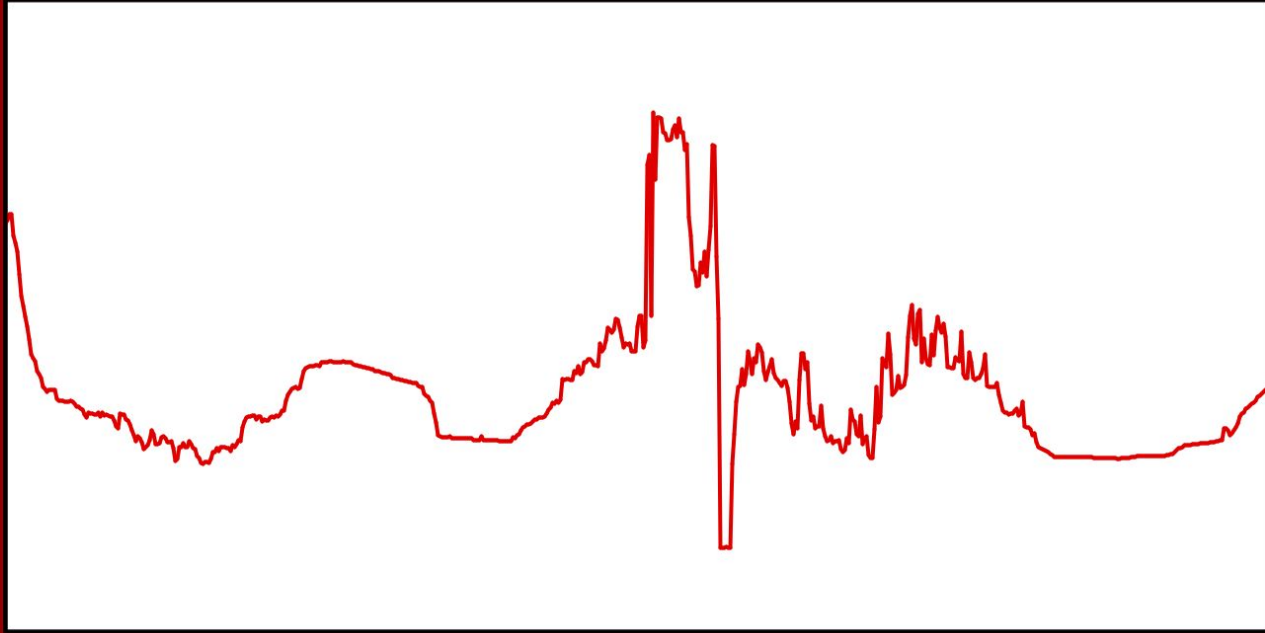


Knobs



MacOS Desktop Icon

Lots And Lots Of UI(Post Graphics Design)



Smpl A Sketch



The screenshot displays the G4P GUI Builder 4.4.2 software interface. The main workspace shows a sketch titled "Smpl A Sketch" with two knobs, "knob1" (labeled "Mode") and "knob2" (labeled "Vol."). The interface includes a top toolbar with various drawing tools, a left sidebar with a "Main window" tab, and a right sidebar with a "Properties" panel. The "Properties" panel shows settings for "GUI LOCK" and "Variable Name" for "knob2", including event methods, initial values, minimum/maximum values, number of ticks, and easing options.

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```

Lots And Lots Of UI(Display Modes+Volume)

```
float[] resetSin = genSinWav();
float[] resetEnv = genBasicEnv();
float prevMode = 0.0;

void wavMode(float[] wav) { //Draw Screen, draw wav, draw outline
  drawScreen();
  drawWav(wav);
  drawScreenBox();
}

void envMode(float[] env) { //Draw Screen, draw env, draw outline
  drawScreen();
  drawEnv(env);
  drawScreenBox();
}

void imgMode() { //Draw Screen, draw wavs(need to add), draw outline
  drawScreen();
  drawScreenBox();
}

void checkMode() { //Would do else ifs but this logic makes the knob smoother
  if (0.0 == mode || prevMode == 0.0) {
    if (clearWav) {
      wavMode(resetSin);
    } else {
      wavMode(processWavImage());
    }
    prevMode = 0.0;
  }
  if (1.0 == mode || prevMode == 1.0) {
    if (clearEnv) {
      envMode(resetEnv);
    } else {
      envMode(getTimeVals());
    }
    prevMode = 1.0;
  }
  if (2.0 == mode || prevMode == 2.0) {
    imgMode();
    prevMode = 2.0;
  }
}
```

```
public void knob2_turn1(GKnob source, GEvent event) { // _CODE_:knob2:873452:
  //println("knob2 - GKnob >> GEvent." + event + " @ " + millis());
  volume = Math.round(source.getValueI()) * 0.1; //Round volume
  s.volume(volume);
} // _CODE_:knob2:873452:
```

Lots And Lots Of UI(Various Bug Fixes)

Removed Download Option

 clarinetgamer committed 1 hour ago

Fixed Old Documentation

 clarinetgamer committed 1 hour ago

Added MacOS Download 4 Progress Check

 clarinetgamer committed 1 hour ago

WE DID IT

 clarinetgamer committed 2 hours ago

Bug Fixes

 clarinetgamer committed 2 hours ago

Cache Handling

 clarinetgamer committed 3 hours ago

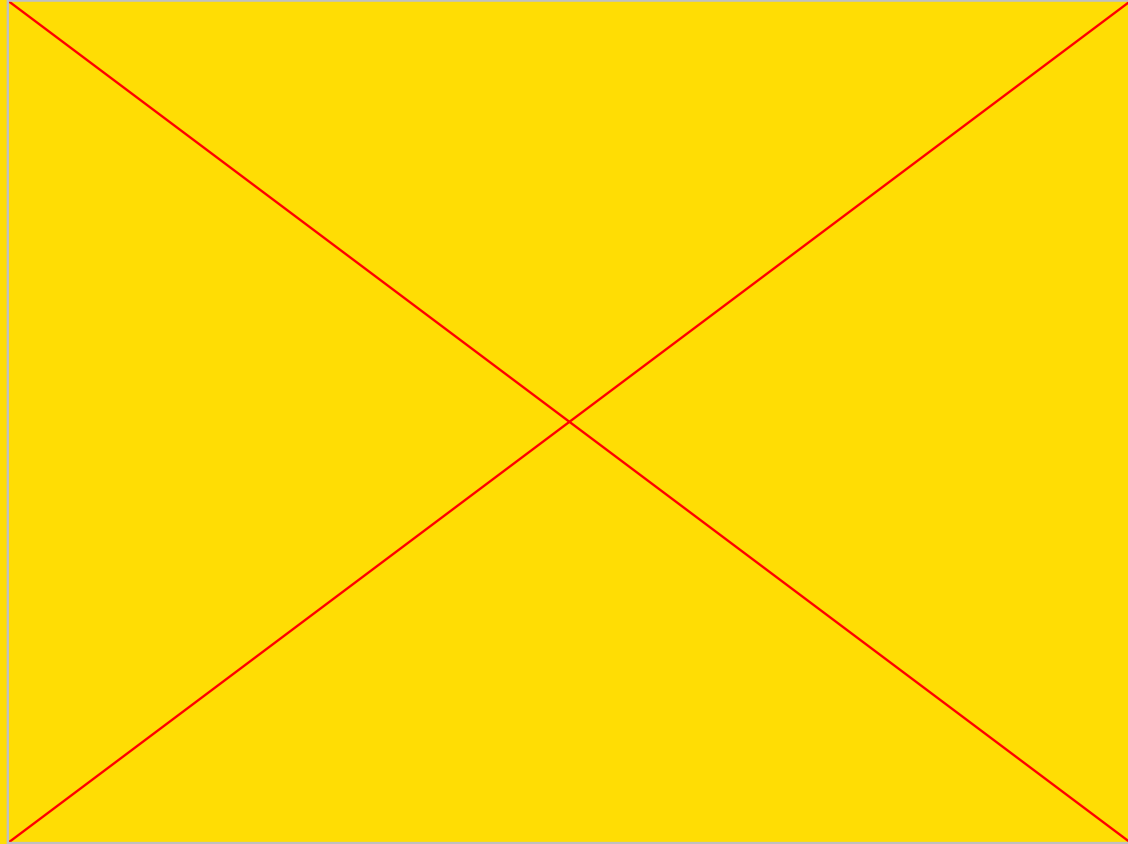
Ascii comunication

 clarinetgamer committed 3 hours ago

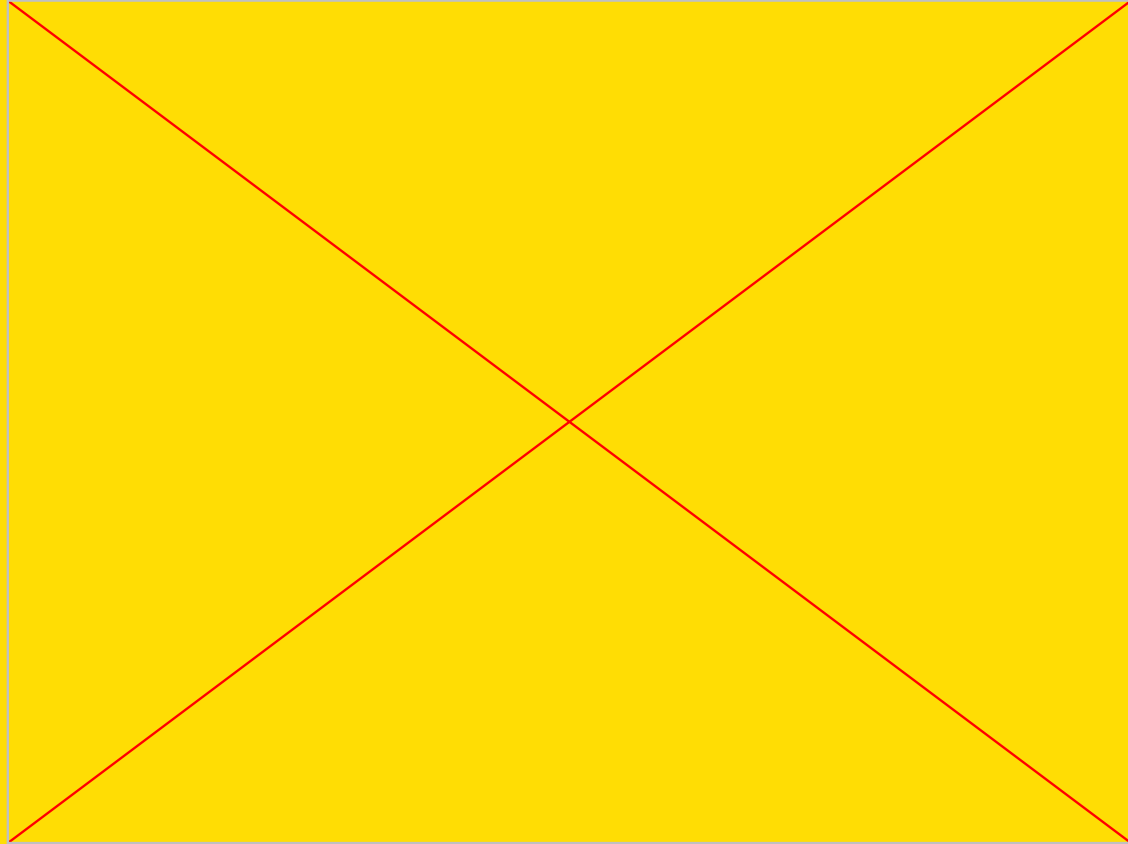
Cookin

 clarinetgamer committed 4 hours ago

Demo Modes



Demo Wav Display



Demo Wav Reset

